

# Optimal Coordination of Water Distribution Energy Flexibility With Power Systems Operation

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**Abstract**—This paper proposes a model for optimizing the energy flexibility of water distribution systems (WDSs) in day-ahead power systems operation. The water distribution system operators (W-DSOs) are considered as energy-conscious entities, who run the proposed WDS operation model to optimize the operation of pumps and tanks for minimizing the operation cost of local WDSs, given the expected water demand and electricity prices of the next day. A WDS energy flexibility model is proposed that is used by W-DSOs in order to calculate and offer the feasible flexible energy capacity to the power system operator. The proposed WDS operation model takes into account the hydraulic operating constraints of water networks, thus ensuring deliverability of the WDS energy flexibility. Further, a network-constrained unit commitment model is proposed that integrates and optimizes the energy flexibility of WDSs in the day-ahead operation of power systems. The proposed model is implemented on a 15-node WDS that is powered by a 6-bus power system and the results are presented to study the impacts of optimized WDS energy flexibility on power and water systems operation.

**Index Terms**—Water distribution systems, energy flexibility, mixed-integer linear programming, unit commitment, demand response, water-energy nexus.

## I. INTRODUCTION

WATER distribution system (WDS) facilities, including water and wastewater treatment plants, are energy-intensive infrastructures that account for about 3–4% of the total electricity used in the United States [1]. Approximately 80% of this total electricity is used for pumping and distributing water and wastewater with the remaining to account for water treatment [2]. Further, drinking water and wastewater utilities are typically the largest energy consumers of municipal governments, accounting between 30–40% of total energy consumed [3]. In California, water facilities consume 20% of the state's total electricity [4], out of which, urban water usage constitutes more than 70% of the electricity associated with water supply and treatment [5]. As population grows, electricity demand associated with water supply and treatment is projected to increase.

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Traditionally, water and power systems have been designed and operated as two uncoupled systems. However, in reality these systems are mutually interdependent [6], [7]. Water is utilized, often in large amounts, in energy sector for mining, fuel production, hydropower, and power plant cooling. On the other hand, energy is an indispensable component of the water facilities as electricity is used for pumping, treatment and distribution of water. This interrelationship is often referred to as water-energy nexus and couples these critical infrastructures upon which human civilization depends [8].

## A. Background on Water-Energy Nexus

Predominantly, the water-energy nexus has been studied with respect to the policy and regulatory challenges and its interactions with the food supply, climate change, economic growth, and water rights [9]–[11]. Recent research has also investigated the impact of climate change on the operation of hydroplants and water reservoir management [12]. From the modeling perspective, the water-energy nexus has been studied using multiple technical and economic approaches, mostly looking at the co-production facilities as the coupling components between the water and power networks [13]–[19]. In [13], the coupling between the power and water systems is studied in terms of utilizing co-production facilities (e.g., hydroelectric and thermal desalination plants) assuming that all the facilities are connected into a single node in the power system. Additional work has also investigated the role of battery storage units in the optimal dispatch of co-production facilities [14]. In [15], the link between water and power systems is studied through the operation of desalination units. Robust optimization has been applied to hydrothermal generation scheduling problem in [16], without modeling constraints of the power transmission networks. Ignoring power transmission network constraints is widely observed in works where the primary scope is to schedule the operation of pumps in water system facilities [17], [18]. In a different approach, the water-energy nexus has been modeled as a physics-based integrated engineering system in [19] where a set of algebraic equations was derived to describe the energy input-output relations between the electricity, water and wastewater systems using bond graph models. Although there has been a recent research focus in regards to investigating and modeling the water-energy nexus [9]–[19], less attention has been given to account for the energy flexibility that water facilities can introduce to the operation of power systems.

The large energy consumption of water facilities along with their flexible components such as water pumps and tanks make them suitable candidates for energy efficiency and optimization applications. A joint effort between power system operators and the water facilities they serve would create a great opportunity for enhancing the flexibility of power systems and increasing the energy efficiency of water industry [20]. A model is developed in [21] for optimizing the demand response and regulation capacity that a WDS can offer in power systems operation. The other efforts in technical literature are mainly focused on passive energy efficiency and price-based load management strategies in water distribution facilities [22]–[24]. From the power system point of view, the energy efficiency programs could improve the system operations, reduce operation costs, and possibly reduce emission from the electricity sector [25], [26]. Several pilot energy efficiency programs that include loads from water facilities are conducted and their economic impacts on power systems operation is studied [27].

The water distribution system operators (W-DSOs) have a natural incentive to act energy-conscious, where the operation coordination with power systems could reduce their electricity costs [8]. However, W-DSOs may be also concerned about the negative impacts of modifying the operation of their systems on the reliable delivery of water to the consumers. This would require a comprehensive model to optimize the dynamic energy flexibility of WDSs considering their underlying hydraulic constraints. However, the majority of technical literature models the water-energy optimization problem at supply side neglecting the power flow limits of power transmission networks and the hydraulic constraints of WDSs. Therefore, the dispatch decisions of these models may not be supported by the constrained power and water network characteristics.

### B. Contribution and Paper Structure

This paper investigates the water-energy nexus in the context of WDS operation, and elaborates WDS modeling as part of integrating the flexibility of its energy-intensive components (pumps) and storage plants (tanks) in power systems operation. More specifically, this paper proposes a model, shown in Fig. 1, to integrate and optimize the energy flexibility of WDSs in the day-ahead operation of power systems. In Fig. 1, a day-ahead WDS operation model is proposed that utilizes the expected water demand and electricity prices of the next day to optimize the operation of water pumps and tanks for minimizing the operation cost of WDSs. In order to ensure the deliverability of the scheduled energy flexibility, the proposed day-ahead WDS operation model incorporates a detailed formulation of the water distribution network and its associated hydraulic constraints. However, the energy flexibility scheduled by W-DSOs may not necessarily be optimal from the power system operator's viewpoint. Therefore, a network-constrained unit commitment model is proposed to co-optimize the energy flexibility of WDS with day-ahead schedule of generating units considering the power flow constraints of the transmission network. The WDS energy

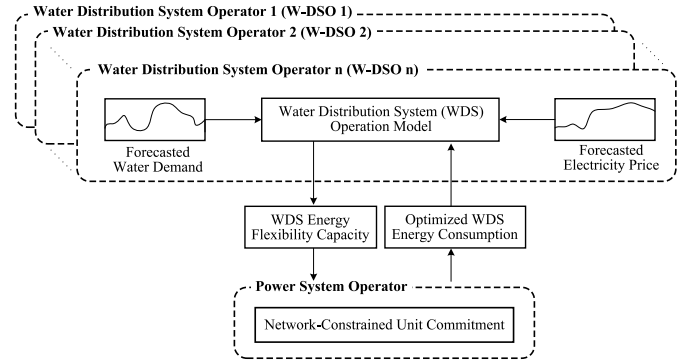


Fig. 1. Model for integrating WDS energy flexibility in power systems.

flexibility is quantified in terms of upper and lower WDS energy flexibility limits by which the feasible capacity range of the WDS energy consumption flexibility is determined. In summary, the main contributions of this paper are as follows:

- This paper develops a model that integrates and optimizes the energy flexibility of W-DSOs in day-ahead power systems operation and ensures the deliverability of the scheduled energy flexibility by considering the hydraulic constraints of the water distribution networks.
- A WDS energy flexibility model is proposed that offers the feasible energy flexibility capacity range of the WDSs to the power system operators. The proposed energy flexibility model enables the power system operators to optimize decisions on utilization of WDS energy flexibility in order to minimize the operation cost of network-constrained power systems.
- In order to ensure the computational efficiency of the WDS operation model for large-scale WDSs, this paper proposes a technique to linearize the nonlinear terms that eventually converts the nonlinear model to a mixed-integer linear programming problem that improves the computational efficiency of the proposed model.

The rest of this paper is organized as follows: the proposed WDS operation model is formulated in Section II as a nonlinear optimization problem. A linearization method is developed in Section III that casts the proposed nonlinear WDS operation model into a mixed-integer linear programming problem. Section IV presents the proposed WDS energy flexibility model, as well as the proposed network-constrained unit commitment model for integrating the WDS energy flexibility in day-ahead power systems operation. The simulation results on a 6-bus power system and 15-node WDS are presented in Section V, and the conclusions are drawn in Section VI.

## II. WATER DISTRIBUTION SYSTEM OPERATION

In this section, we present the proposed model to optimize the operation of W-DSOs. The proposed WDS operation optimization problem models the components of the water distribution network (including pipes, water pumps, water tanks, reservoirs and water loads) and its associated hydraulic constraints in detail. The schematic diagram of WDS and the associated hydraulic components are illustrated in Fig. 2. From a mathematical point of view, a WDS is represented by a

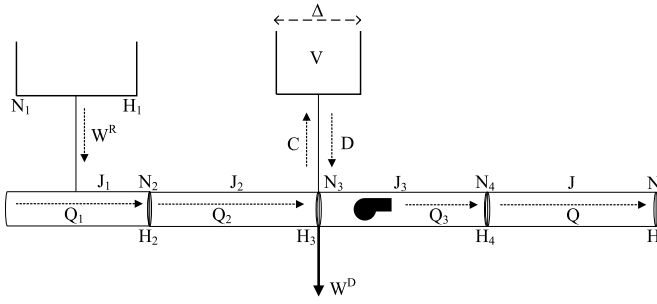


Fig. 2. Schematic diagram of water distribution system components.

directed graph  $G = (\mathcal{N}, \mathcal{J})$ , where  $\mathcal{N}$  is the set of  $N$  nodes representing  $R$  reservoirs and  $T$  tanks, and  $\mathcal{J}$  is the set of  $J$  arcs comprising  $S$  pipes and  $U$  water pumps. The pressure head of the nodes is represented by  $N \times 1$  decision variable vector  $\mathbf{H}_t = [H_{t,1}, \dots, H_{t,N}]^T$ , and the volumetric flow rate of arcs (pipes and pumps) is shown by  $J \times 1$  decision variable vector  $\mathbf{Q}_t = [Q_{t,1}, \dots, Q_{t,J}]^T$ , where  $N$  and  $J$  respectively represent the total number of nodes and arcs. Water flow through the network is induced by different pressure levels at the nodes. This pressure head difference is described by  $\mathbf{A}^T \mathbf{H}_t$ , where  $\mathbf{A}$  is the  $N \times J$  node-arc incidence matrix with rows and columns corresponding to nodes and arcs of the water network, respectively. The flow rate in a pipe can be maintained either through sufficient elevation difference between two nodes or by generating a pressure head by pumping. For example, if a WDS is fed via a ground water supply system so that gravity does not suffice, then a pumping station is required to raise certain pressure node potentials.

#### A. Formulation of the Proposed WDS Operation Model

The proposed WDS operation model is formulated here as a nonlinear optimization problem, where the objective is to minimize the total WDS operation cost (1), including the cost of purchasing water and the cost of energy [28], [29], over the scheduling horizon divided to  $NT$  intervals:

$$\min \sum_{t=1}^{NT} (\mathbf{K}^T \boldsymbol{\Theta} \mathbf{W}_t^R + \mathbf{1}_U^T \mathbf{P}_t^W \pi_{b,t}). \quad (1)$$

The first term in (1) is the cost incurred by purchasing water fed into the network, where  $\mathbf{W}_t^R$  is the  $N \times 1$  vector of water inflows at the reservoir nodes at time  $t$ ,  $\boldsymbol{\Theta}$  is the  $R \times N$  reservoir-node incidence matrix that maps the water reservoirs to nodes, and  $\mathbf{K}$  is the  $R \times 1$  vector of reservoirs' water price. The second term in (1) is the cost of energy needed to operate the pumps, where  $\pi_{b,t}$  is the forecasted electricity price at power transmission bus  $b$  that supplies the WDS's energy, and  $\mathbf{P}_t^W$  is the  $U \times 1$  decision variable vector of pumps' electricity consumption at time  $t$ .

The objective function (1) is constrained to the water distribution flow and water tank and pumps constraints (2)-(16), described in detail below.

1) *Water Conservation Constraint*: The WDS delivers the water from reservoirs to the consumers through a network of pipes, water tanks and pumps. The water conservation

constraint in the WDS is formulated as:

$$\mathbf{W}_t^R - \mathbf{W}_t^D = \mathbf{A} \mathbf{Q}_t + \mathbf{\Lambda}^T [\mathbf{C}_t - \mathbf{D}_t] \quad \forall t, \quad (2)$$

where  $\mathbf{W}_t^D$  is  $N \times 1$  vector of nodes' water demand parameter at time  $t$ ;  $\mathbf{C}_t$  and  $\mathbf{D}_t$  are respectively the  $T \times 1$  vectors of the water inflow and outflow decision variables of tanks at time  $t$ ; and  $\mathbf{\Lambda}$  is  $T \times N$  tank-node incidence matrix. The water inflow and outflow of tanks are explicitly modeled in (2), so the W-DSO may utilize the water storage capacity of tanks to store water and serve water demand in future times.

2) *Water Tank Constraints*: The dynamic operation constraints of water tanks are formulated as:

$$\mathbf{V}_{t+1} = \mathbf{V}_t + \mathbf{C}_t - \mathbf{D}_t \quad \forall t, \quad (3)$$

$$\underline{\mathbf{V}} \leq \mathbf{V}_t \leq \bar{\mathbf{V}} \quad \forall t, \quad (4)$$

$$\mathbf{V}_1 = \mathbf{V}^{init}, \quad (5)$$

$$\mathbf{V}_t = \mathbf{\Delta} \mathbf{\Lambda} \mathbf{H}_t \quad \forall t, \quad (6)$$

$$0 \leq \mathbf{C}_t \leq \bar{\mathbf{C}} \quad \forall t, \quad (7)$$

$$0 \leq \mathbf{D}_t \leq \bar{\mathbf{D}} \quad \forall t, \quad (8)$$

where the dynamic operation of tanks is governed by the state equation (3). The water stored in tanks at time  $t$  is shown by  $T \times 1$  vector  $\mathbf{V}_t$ , which is constrained to the minimum and maximum capacity limits  $\underline{\mathbf{V}}$  and  $\bar{\mathbf{V}}$  in (4). Constraint (5) sets the water volume stored in tanks at the start of scheduling horizon, where  $\mathbf{V}^{init}$  is the vector of initial stored water in tanks. The water stored in tanks drives the pressure head at the tank nodes in (6), where  $\mathbf{\Delta}$  is the  $T \times T$  diagonal matrix of tanks surface parameters. The water inflow and outflow rate of tanks is also limited in (7) and (8), where  $\bar{\mathbf{C}}$  and  $\bar{\mathbf{D}}$  are the maximum inflow and outflow rates, respectively.

3) *Pressure Head Loss Constraints*: The nodal pressure head  $\mathbf{H}_t$  is modified by pressure loss in pipes. The pressure loss in pipes due to friction is described by the empirical *Hazen-Williams* head loss constraint (9), [30], [31], where  $\mathbf{G}$  is the  $S \times S$  diagonal matrix of the pipe resistance parameters, and  $\Pi(\cdot)$  is the unit sign function. The nodal pressure heads are constrained in (10) to the minimum and maximum pressure levels  $\underline{\mathbf{H}}$  and  $\bar{\mathbf{H}}$  at the nodes. These bounds are specific for each WDS and depend on pipes' design specifications. The nodal pressure heads are fixed at the reservoir nodes in (11), assuming that the water pressure is fixed at source nodes by a fixed geographical height, where  $\mathbf{Z}$  is  $R \times 1$  vector of reservoir nodes' elevation. The volumetric flow rate decision variable of arcs,  $\mathbf{Q}_t$ , is bounded in (12), where the bound  $\bar{\mathbf{Q}}$  depends on the pipes' diameter and pumps' size.

$$-\mathbf{A}^T \mathbf{H}_t = \mathbf{G} \Pi(\mathbf{Q}_t) |\mathbf{Q}_t|^{1.852} \quad \forall t, \quad (9)$$

$$\underline{\mathbf{H}} \leq \mathbf{H}_t \leq \bar{\mathbf{H}} \quad \forall t, \quad (10)$$

$$\boldsymbol{\Theta} \mathbf{H}_t = \mathbf{Z} \quad \forall t, \quad (11)$$

$$-\bar{\mathbf{Q}} \leq \mathbf{Q}_t \leq \bar{\mathbf{Q}} \quad \forall t. \quad (12)$$

4) *Water Pump Constraints*: Water pumps are active components in WDS that control the pressure of water flowing through them. The increase in water pressure produced by pumps, as well as the power consumption of pumps are formulated in (13) and (14) respectively, where  $\boldsymbol{\Omega}_t$  is the  $U \times 1$

decision variable vector of pump speed at time  $t$ ,  $\mathbf{B}$  is  $U \times J$  pump-arc incidence matrix,  $\mathbf{1}^U$  is a  $U \times 1$  vector of ones, and  $a, b, c, d$  and  $e$  are pump-specific parameters [30], [31].<sup>1</sup> In (15), the pump power consumption is limited to its maximum capacity,  $\mathbf{P}^W$ , and (16) ensures that the water flow rate in pumps is always positive as pumps can only work in one direction.

$$\mathbf{B}\mathbf{A}^T\mathbf{H}_t = \mathbf{\Omega}_t^2 \odot (a\mathbf{1}^U - b((\mathbf{1}^U \oslash \mathbf{\Omega}_t) \odot \mathbf{B}\mathbf{Q}_t)^c) \quad \forall t, \quad (13)$$

$$\mathbf{P}_t^W = \mathbf{\Omega}_t^3 \odot (d\mathbf{1}^U - e((\mathbf{1}^U \oslash \mathbf{\Omega}_t) \odot \mathbf{B}\mathbf{Q}_t)) \quad \forall t, \quad (14)$$

$$0 \leq \mathbf{P}_t^W \leq \mathbf{P}^W \quad \forall t, \quad (15)$$

$$\mathbf{B}\mathbf{Q}_t \geq 0 \quad \forall t. \quad (16)$$

In summary, (1)-(16) present the proposed operation model for W-DSOs, which minimizes the total WDS operation cost in (1) by optimizing the operation of tanks and pumps while ensuring the deliverability of water to the consumers considering the water conservation and pressure head constraints.

### III. LINEARIZATION OF THE NONLINEAR FUNCTION

The proposed WDS operation model (1)-(16) is a nonlinear programming (NLP) problem, with the nonlinearities represented by the pressure head loss constraint (9), and the water pumps' operation constraints (13) and (14). Solution of the NLP model, however, may be time-intensive and complex in large WDSs, and a feasible global optimal solution may not be always guaranteed. In order to address this issue, this section develops a model to approximate the nonlinear functions in (9), (13) and (14) with piecewise linear functions.

#### A. Linearization of the Pressure Head Loss Function

Let  $L_{s,t}(Q_{s,t}) = H_{n,t} - H_{n+1,t}$  shows the time-dependent pressure head loss in pipe  $s$  connecting two nodes  $n$  and  $n+1$  in the water network, where  $Q_{s,t}$  is the water flow rate of the pipe at time  $t$ . Using (9),  $L_{s,t}(Q_{s,t})$  is a nonlinear function of  $Q_{s,t}$  as  $L_{s,t}(Q_{s,t}) = G_s \Pi(Q_{s,t}) |Q_{s,t}|^{1.852}$ . A hypothetical nonlinear pressure head loss function is illustrated in Fig. 3.

In order to develop the piecewise linear approximation of the univariate function  $L_{s,t}(Q_{s,t})$ , let us divide the axis  $Q_{s,t}$  over the range of the function to  $K-1$  segments using  $k = 1, 2, \dots, K$  sampling coordinates (breakpoints)  $q_s^1, q_s^2, \dots, q_s^K$ . The values  $q_s^1$  and  $q_s^K$  coincide with the lower and upper bounds of the  $Q_{s,t}$  domain. As shown in Fig. 3, each breakpoint  $k$  is associated with a continuous variable  $\lambda_{s,t}^k \in [0, 1]$ , and each  $k^{\text{th}}$  interval  $[q_s^k, q_s^{k+1}]$  is associated with a binary variable  $\alpha_{s,t}^k$ .

Any given point  $q_s^k \leq Q_{s,t} \leq q_s^{k+1}$  can be uniquely represented by (17), as a linear combination of two consecutive breakpoints weighted by the associated variables  $\lambda_{s,t}^k$ . The nonlinear function  $L_{s,t}(Q_{s,t})$  is then approximated by linear function  $\hat{L}_{s,t}$  in (18) as convex combination of the function values evaluated at the breakpoints weighted by the associated variables  $\lambda_{s,t}^k$ . Constraint (19) ensures that only one binary

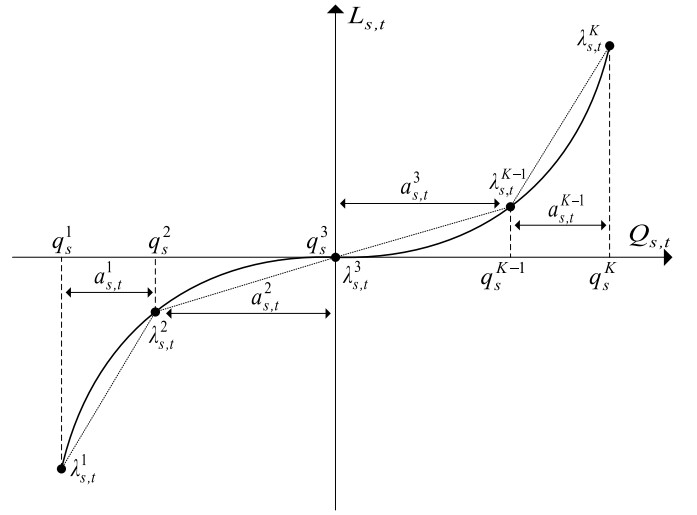


Fig. 3. Piecewise linear approximation of pressure head loss function.

variable  $\alpha_{s,t}^k$  can take the value 1, and thus only one line segment is active at each time. Accordingly, constraints (20)-(24) ensure that only a pair of consecutive positive continuous variables,  $\lambda_{s,t}^k$  and  $\lambda_{s,t}^{k+1}$  take non-zero values.

$$Q_{s,t} = \sum_{k=1}^K q_s^k \lambda_{s,t}^k \quad \forall t, \forall s, \quad (17)$$

$$\hat{L}_{s,t} = \sum_{k=1}^K L_{s,t}(q_s^k) \lambda_{s,t}^k \quad \forall t, \forall s, \quad (18)$$

$$\sum_{k=1}^{K-1} \alpha_{s,t}^k = 1 \quad \forall t, \forall s, \quad (19)$$

$$\sum_{k=1}^K \lambda_{s,t}^k = 1 \quad \forall t, \forall s, \quad (20)$$

$$\lambda_{s,t}^k \leq \alpha_{s,t}^{k-1} + \alpha_{s,t}^k \quad \forall t, \forall s, k = 2, \dots, K-1, \quad (21)$$

$$\lambda_{s,t}^1 \leq \alpha_{s,t}^1 \quad \forall t, \forall s, \quad (22)$$

$$\lambda_{s,t}^K \leq \alpha_{s,t}^{K-1} \quad \forall t, \forall s, \quad (23)$$

$$0 \leq \lambda_{s,t}^k \leq 1 \quad \forall t, \forall s, \forall k. \quad (24)$$

#### B. Linearization of the Water Pump Constraints

Let function  $\Delta h_{u,t}(Q_{u,t}, \Omega_{u,t}) = H_{n+1,t} - H_{n,t}$  shows the time-dependent pressure head increase produced by pump  $u$  connected between the nodes  $n$  and  $n+1$  in the water network, where  $Q_{u,t}$  and  $\Omega_{u,t}$  are respectively the water flow rate and speed of the pump at time  $t$ . Using (13),  $\Delta h_{u,t}(Q_{u,t}, \Omega_{u,t})$  is a nonlinear function defined as  $\Delta h_{u,t}(Q_{u,t}, \Omega_{u,t}) = \Omega_{u,t}^2 (a - b(\frac{1}{\Omega_{u,t}} Q_{u,t})^c)$ . In addition, the pump power consumption is a nonlinear function defined in (14) as  $P_{u,t}^W(Q_{u,t}, \Omega_{u,t}) = \Omega_{u,t}^3 (d - e(\frac{1}{\Omega_{u,t}} Q_{u,t}))$ .

The linearization method described in Section III-A can be extended to linearize the bivariate functions  $\Delta h_{u,t}(Q_{u,t}, \Omega_{u,t})$  and  $P_{u,t}^W(Q_{u,t}, \Omega_{u,t})$ . Let us divide the axis  $Q_{u,t}$  and  $\Omega_{u,t}$  respectively to  $V-1$  and  $M-1$  segments using breakpoints  $q_u^1, q_u^2, \dots, q_u^V$  and  $\omega_u^1, \omega_u^2, \dots, \omega_u^M$ . The values  $q_u^1$  and  $q_u^V$  as

<sup>1</sup> Signs  $\odot$  and  $\oslash$  in (13) and (14) denote respectively the component-wise matrix multiplication and division.

well as  $\omega_u^1$  and  $\omega_u^M$  coincide with the lower and upper bounds of the  $Q_{u,t}$  and  $\Omega_{u,t}$  domains [32].

The segments on the axes partition the bivariate function domain into several rectangles. Consider the rectangle defined by vertices  $(q_u^v, \omega_u^m)$ ,  $(q_u^{v+1}, \omega_u^m)$ ,  $(q_u^{v+1}, \omega_u^{m+1})$ ,  $(q_u^v, \omega_u^{m+1})$  on the function domain (see Fig. 4). This rectangle is covered by two triangles defined by vertices  $(q_u^v, \omega_u^m)$ ,  $(q_u^v, \omega_u^{m+1})$ ,  $(q_u^{v+1}, \omega_u^{m+1})$ , and  $(q_u^v, \omega_u^m)$ ,  $(q_u^{v+1}, \omega_u^m)$ ,  $(q_u^{v+1}, \omega_u^{m+1})$ , and divided by the diagonal  $(q_u^v, \omega_u^m)$ ,  $(q_u^{v+1}, \omega_u^{m+1})$ . A continuous variable  $\gamma_{u,t}^{v,m} \in [0, 1]$  is associated with each vertex  $(v, m)$ , and binary variables  $\bar{h}_{u,t}^{v,m}$  ( $v = 1, \dots, \mathcal{V} - 1; m = 2, \dots, M$ ) and  $\underline{h}_{u,t}^{v,m}$  ( $v = 2, \dots, \mathcal{V}; m = 1, \dots, M - 1$ ) are respectively associated with the upper left and lower right triangles.

Any given pump operating point  $q_u^v \leq Q_{u,t} \leq q_u^{v+1}$  and  $\omega_u^m \leq \Omega_{u,t} \leq \omega_u^{m+1}$  can be uniquely represented by (25) and (26), as linear combinations of the vertex coordinates weighted by the associated variables  $\gamma_{u,t}^{v,m}$ . The bivariate nonlinear functions  $\Delta h_{u,t}(Q_{u,t}, \Omega_{u,t})$  and  $P_{u,t}^W(Q_{u,t}, \Omega_{u,t})$  are then approximated by linear functions  $\hat{\Delta}h_{u,t}$  and  $\hat{P}_{u,t}^W$  in (27) and (28), as convex combination of the function values evaluated at the vertices of the triangle containing the operating point  $(Q_{u,t}, \Omega_{u,t})$ , weighted by the associated variables  $\gamma_{u,t}^{v,m}$ . Constraint (29) ensures that in any solution only one of the triangles is chosen. Constraints (30)-(32) ensure that in any solution at most three of continuous variables  $\gamma_{u,t}^{v,m}$  can take non-zero values.

$$Q_{u,t} = \sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M q_u^v \gamma_{u,t}^{v,m} \quad \forall t, \forall u, \quad (25)$$

$$\Omega_{u,t} = \sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M \omega_u^m \gamma_{u,t}^{v,m} \quad \forall t, \forall u, \quad (26)$$

$$\hat{\Delta}h_{u,t} = \sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M \Delta h_{u,t}(q_u^v, \omega_u^m) \gamma_{u,t}^{v,m} \quad \forall t, \forall u, \quad (27)$$

$$\hat{P}_{u,t}^W = \sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M P_{u,t}^W(q_u^v, \omega_u^m) \gamma_{u,t}^{v,m} \quad \forall t, \forall u, \quad (28)$$

$$\sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M (\bar{h}_{u,t}^{v,m} + \underline{h}_{u,t}^{v,m}) = 1 \quad \forall t, \forall u, \quad (29)$$

$$\sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M \gamma_{u,t}^{v,m} = 1 \quad \forall t, \forall u, \quad (30)$$

$$\gamma_{u,t}^{v,m} \leq \bar{h}_{u,t}^{v,m-1} + \bar{h}_{u,t}^{v+1,m} + \bar{h}_{u,t}^{v,m} + \underline{h}_{u,t}^{v-1,m} + \underline{h}_{u,t}^{v,m+1} + \underline{h}_{u,t}^{v,m} \quad \forall t, \forall u, \forall v, \forall m, \quad (31)$$

$$0 \leq \gamma_{u,t}^{v,m} \leq 1 \quad \forall t, \forall u, \forall v, \forall m. \quad (32)$$

In summary, the linear model in Section III-A approximates the nonlinear function in (9), and the linear models in Section III-B approximate the nonlinear functions in (13) and (14). Substituting the nonlinear functions with the associated linear approximations converts the proposed nonlinear WDS operation model (1)-(16) into a linear model. The proposed linear models in this section introduce new continuous and binary variables to the problem, so the linearized

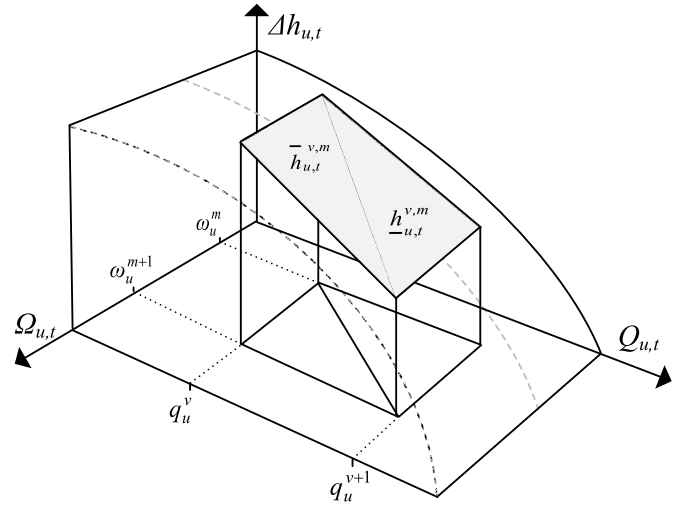


Fig. 4. Geometric representation of the convex combination triangulation method applied to the pump pressure head increase function.

model is an instance of mixed-integer linear programming (MILP) problem.

#### IV. WDS ENERGY FLEXIBILITY OPTIMIZATION

The WDS operation model proposed in Section II would allow the W-DSOs to minimize the operation cost of their local WDSs, while supplying the water demand at all times. Consider a power system that supplies the power requirement of  $ND$  water distribution systems. Let  $P_{d,t}^{W,opt}$  and  $P_{d,t}^{W,nopt}$  be the total hourly power consumption of water distribution system  $d \in ND$ , with and without optimizing the operation of flexible components using the proposed model. The optimized energy consumption of W-DSO  $d$ ,  $P_{d,t}^{W,opt}$ , is calculated using the W-DSO operational model (1)-(16), where the nonlinear constraints could be substituted by the linear ones formulated in Section III. In addition, the non-optimized power consumption of W-DSO  $d$ ,  $P_{d,t}^{W,nopt}$ , is calculated by removing the energy cost term from objective function of the proposed W-DSO operation problem, where the problem would only optimize the WDS operation to minimize the water procurement costs. The optimized and non-optimized WDS power consumption profiles would respectively provide the lower and upper bounds for the WDS power consumption and the associated costs. The upper and lower bounds of the WDS power consumption provide the feasible range of WDS energy flexibility that could be offered to power systems. In this section, we develop a model to integrate and optimize the energy flexibility of water distribution systems in power systems operation.

##### A. WDS Energy Flexibility Model

The proposed WDS energy flexibility model is presented in (33)-(37). In (33),  $P_{d,t}^W$  is the total hourly power consumption of WDS  $d$ , which is equal to the non-optimized power consumption,  $P_{d,t}^{W,nopt}$ , plus the flexible energy capacity of the WDS,  $P_{d,t}^{W,f}$ , utilized by the power system operator. The utilized flexible energy  $P_{d,t}^{W,f}$  is a continuous variable that is

limited in (34) to the flexible energy capacity bounds. The bounds are defined in (35) and (36), as the difference between the optimized and non-optimized power consumption of the WDSs. In (35), when the difference is negative, the minimum energy flexibility bound is defined by this negative value, otherwise it is zero. Similarly, in (36), when the difference is positive, the maximum energy flexibility bound is defined by this positive value, otherwise it is zero. A positive energy flexibility value indicates the potential electricity consumption needed to pump and store water in tanks for serving the water demand in future times. On the contrary, a negative energy flexibility indicates the opportunity to reduce the pumps' load and instead using the water stored in the tanks. While the utilized energy flexibility at each time is capped by the available capacity at that time, intertemporal constraint (37) ensures that the pumps would operate earlier in the operating horizon before reducing their load in subsequent hours.

$$P_{d,t}^W = P_{d,t}^{W,opt} + P_{d,t}^{W,f} \quad \forall d, \forall t, \quad (33)$$

$$\underline{P}_{d,t}^{W,f} \leq P_{d,t}^{W,f} \leq \overline{P}_{d,t}^{W,f} \quad \forall d, \forall t, \quad (34)$$

$$\underline{P}_{d,t}^{W,f} = \min [P_{d,t}^{W,opt} - P_{d,t}^{W,nopt}, 0] \quad \forall d, \forall t, \quad (35)$$

$$\overline{P}_{d,t}^{W,f} = \max [P_{d,t}^{W,opt} - P_{d,t}^{W,nopt}, 0] \quad \forall d, \forall t, \quad (36)$$

$$\sum_{t=1}^{NT} P_{d,t}^{W,f} = 0 \quad \forall d. \quad (37)$$

### B. Integration in Power System Operation

As shown in Fig. 1, the power system operator co-optimizes the final decisions on WDS energy consumption with the power system operation, integrating the WDS energy flexibility model in (33)-(37). This is done through the proposed network-constrained unit commitment model formulated as:

$$\min \sum_{t=1}^{NT} \sum_{i=1}^{NG} (C(I_{i,t}, P_{i,t}) + SU_{i,t} + SD_{i,t}), \quad (38)$$

$$\sum_{i=1}^{NG_b} P_{i,t} - P_{b,t}^D - \sum_{d=1}^{ND_b} P_{d,t}^W = \sum_{l=1}^{NL_b} F_{l,t} \quad \forall b, \forall t, \quad (39)$$

$$F_{l,t} = \frac{1}{X_l} (\theta_{l,t}^S - \theta_{l,t}^R) \quad \forall l, \forall t, \quad (40)$$

$$-\overline{F}_{l,t} \leq F_{l,t} \leq \overline{F}_{l,t} \quad \forall l, \forall t. \quad (41)$$

The objective of the proposed model is to minimize the total operation cost of power systems in (38), where  $C(I_{i,t}, P_{i,t})$ ,  $SU_{i,t}$ ,  $SD_{i,t}$  are respectively the generation cost function, startup and shutdown cost of generating unit  $i$  at time  $t$ , and  $NG$  is the number of generating units. The objective function (38) is subject to the power transmission network constraints (39)-(41). The DC power flow equation (39) integrates the power consumption of WDSs in the nodal power balance constraints, where  $P_{b,t}^D$  is power demand of bus  $b$  at time  $t$  excluding the power demand of WDSs in that bus, and  $F_{l,t}$  is the power flow of transmission line  $l$  at time  $t$  that is defined in (40). The power flow of lines are limited to the line capacity bounds in (41). Note that the power consumption of W-DSO  $d$ ,  $P_{d,t}^W$  in (39), is constrained to the proposed WDS energy

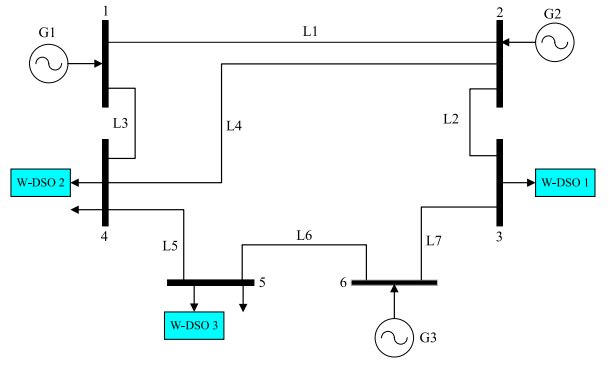


Fig. 5. Online diagram of the 6-bus system.

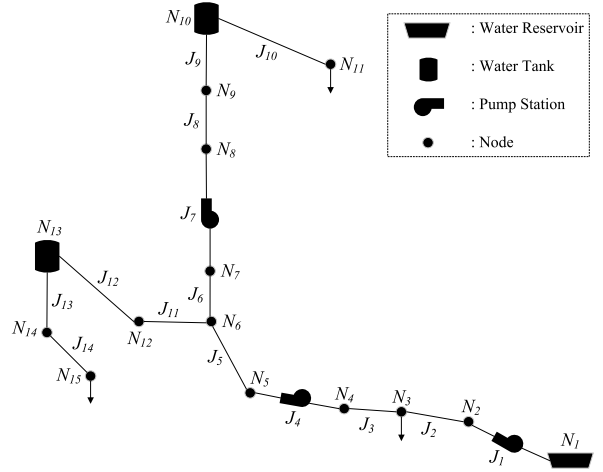


Fig. 6. Schematic representation of the test WDS.

flexibility model in (33)-(37). The power system operation problem is also constrained to the prevailing generating units constraints (i.e., capacity, ramping, minimum on/off time, and startup/shutdown cost). This paper does not intend to repeat the well-understood constraints of generating units, and refer the readers to the literature on detailed MILP formulation of the networked-constrained unit commitment problem [33]–[36].

### V. STUDY RESULTS

The proposed model to integrate the energy flexibility of WDSs in power systems operation is implemented on the 6-bus test power system shown in Fig. 5. The 6-bus system supplies the energy consumption of three identical test WDSs connected to the three load buses 3, 4 and 5. The test WDS, shown in Fig. 6, includes 15 nodes, 14 arcs (including 11 pipes and 3 pumps in arcs 1, 4, 8), 2 tanks in nodes 10, 13, three water load junctions (nodes 3, 11, 15) and one reservoir (node 1) representing the only water source to the system. The data for the 6-bus power system and the test WDS are respectively available in [35] and [37]. The initial water volume of the tanks is set to zero.

Three cases are studied here. In Case 1, the three WDSs are operated without optimizing their energy consumption. In this case, the energy cost term is excluded from (1), so the

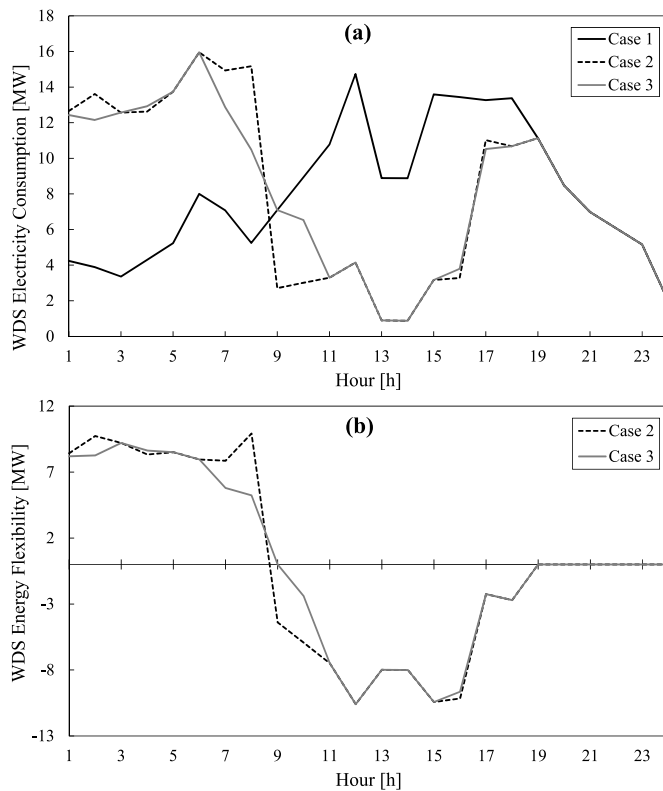


Fig. 7. (a) Daily power consumption of W-DSOs, and (b) energy flexibility schedule of W-DSOs.

W-DSOs' operation goal is to supply water demand while minimizing the cost of purchasing water from the reservoir.

In Cases 2 and 3, the energy consumption of WDSs is optimized using the model proposed in Sections II and III. Case 2 and 3 differ in terms of the method for integrating the available WDS energy flexibility in the power system operation. In Case 2, an uncoordinated WDS operation is considered in which the W-DSOs locally optimize their system operation and submit their adjusted fixed energy consumption to the power system operator. In Case 3, the W-DSOs calculate and submit their energy flexibility to the power system operator, as proposed in Section IV, and the power system operator optimizes and coordinates the available WDS energy flexibility with the power system operation. The numerical results for the three cases are discussed below.

**WDS Energy Flexibility and Consumption Schedule:** The electricity consumption of the WDSs over the one day scheduling horizon in three cases is shown in Fig. 7.(a). In Fig. 7.(a), the WDS energy consumption in Cases 2 and 3 differs from that of Case 1 where the energy consumption is not optimized. In Case 2, the W-DSOs pump water to the storage tanks during off-peak hours 1-8 when the electricity price is lower, and release the water during the peak hours 9-19. The available WDSs energy flexibility in Case 2 is calculated by subtracting the optimized energy consumption of WDSs from the non-optimized energy consumption in Case 1. In Case 2, where the energy flexibility is not optimized by the power system operator, the total available WDS energy flexibility, shown in Fig. 7.(b), is utilized by the W-DSOs.

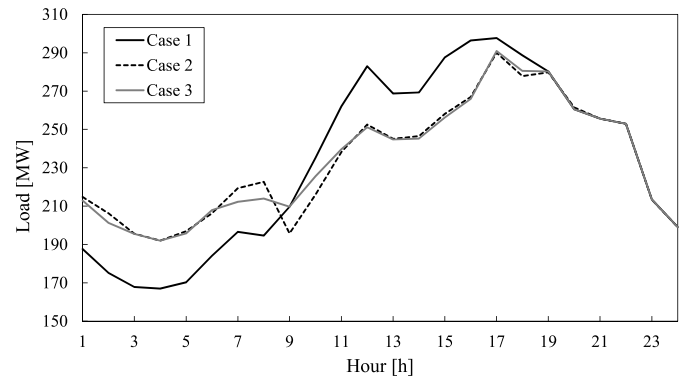


Fig. 8. Daily load profile of the 6-Bus power system.

TABLE I  
OPERATION COSTS OF THE 6-BUS POWER SYSTEM

| Case | Power System Operation Cost (\$) |
|------|----------------------------------|
| 1    | 92,868.7                         |
| 2    | 89,228.1                         |
| 3    | 88,121.0                         |

In Case 3, the power system operator optimizes the utilization of WDSs energy flexibility, within the WDSs energy flexibility bounds. As shown in Fig. 7.(b), the optimized energy flexibility in Case 3 is not equal to the available flexibility in Case 2, which means that the power system operator does not necessarily utilize the total available WDS energy flexibility. As a result, the energy consumption of WDSs in Case 3 is different from Case 2 in Fig. 7.(a).

**Power System Load Profile:** The total load of the 6-bus system for the study cases is shown in Fig. 8. In Fig. 8, optimizing the energy consumption of WDSs in Cases 2 and 3 has reduced the peak load of the 6-bus system, as compared to Case 1 where the energy consumption of WDSs is not optimized. In Case 1, all the three generating units of the 6-bus system are committed to supply the daily load profile. However, the more expensive unit G2 is only committed for two hours 17-19 in Cases 2 and 3 where the energy flexibility of WDSs reduces the peak load of the system.

**Power System Operation Cost:** The total operation cost of the 6-Bus system in the three study cases is shown in Table I. In Table I, the operation cost is reduced to \$89,228.1 in Case 2 as compared to Case 1, which shows the operational cost benefits to the power system in the presence of WDSs energy flexibility. The operating cost in Case 3, where the WDS energy flexibility is optimized by the power system operator, is further reduced to \$88,121.0, which demonstrates a higher cost saving than Case 2. This result emphasizes that the WDS energy flexibility would deliver additional benefits to power systems operation when it is optimized by the power system operator.

Integrating the WDS energy flexibility in the 6-bus system operation also modifies the locational marginal prices (LMPs). The average LMPs of the 6-bus system buses are shown in Fig. 9. In Fig. 9, utilization of the WDS energy flexibility in Cases 2 and 3 reduces the LMPs during the off-peak hours

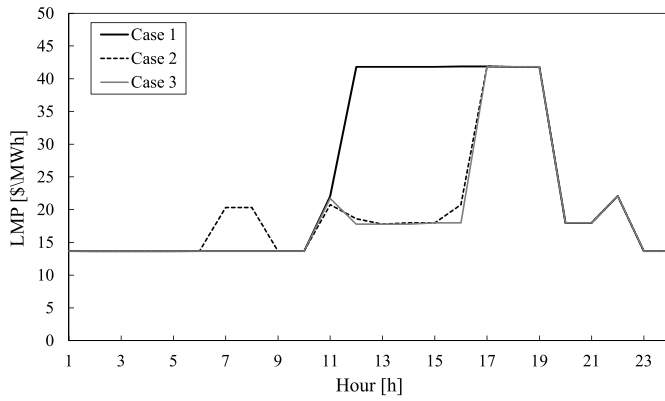


Fig. 9. Average LMPs of the 6-bus system buses.

TABLE II  
W-DSOs ENERGY COST

| W-DSO   | W-DSO Daily Energy Cost (\$) |          |          |
|---------|------------------------------|----------|----------|
|         | Case 1                       | Case 2   | Case 3   |
| W-DSO 1 | 5,761.4                      | 4,208.2  | 3,841.9  |
| W-DSO 2 | 7,239.3                      | 4,892.8  | 4,400.4  |
| W-DSO 3 | 6,958.9                      | 4,773.7  | 4,294.4  |
| Total   | 19,959.6                     | 13,874.7 | 12,536.8 |

12-17. The reason for this is that the expensive unit G2 is not committed in these hours. The results in Fig. 9 show that the LMP curve is smoother in Case 3 where the WDS energy flexibility is optimized by the power system operator.

**Water Distribution System Energy Cost:** The proposed W-DSO optimization model applied in Cases 2 and 3 results in reducing the energy consumption cost for the WDSs in Table II. The W-DSOs' energy cost in Table II are calculated by multiplying the W-DSOs energy consumption in each case with the associated locational marginal prices (LPMs). In Table II, energy cost of all three W-DSOs is reduced in Case 3, which demonstrates a total of 37.1% cost savings as compared to Case 1. In Table II, more energy cost savings are observed for Case 3 than for Case 2. This result shows that the W-DSOs would benefit more when their energy flexibility is optimized by the power system operator. In Table II, more energy cost savings are achieved for W-DSO 2 and W-DSO 3 than for W-DSO 1, which shows that the energy optimization of WDSs is more attractive, financially, in power system buses 4 and 5. This result also highlights the fact that the energy costs of WDSs are linked to the operational characteristics of power systems. The cost savings in Table II are based on a single-day operation which would be added up on an annual basis.

#### A. Comparison of Linear and Nonlinear W-DSO Models

The goal here is to compare the performance of the proposed linear W-DSO operation model in Section III with the nonlinear counterpart presented in Section II. In this regard, the operation of the test WDS, shown in Fig. 6, is simulated using both the MILP and NLP models, which are respectively solved using CPLEX and COUENNE solvers in GAMS environment, on a desktop computer with a 4.0-GHz i7 processor and 32 GB

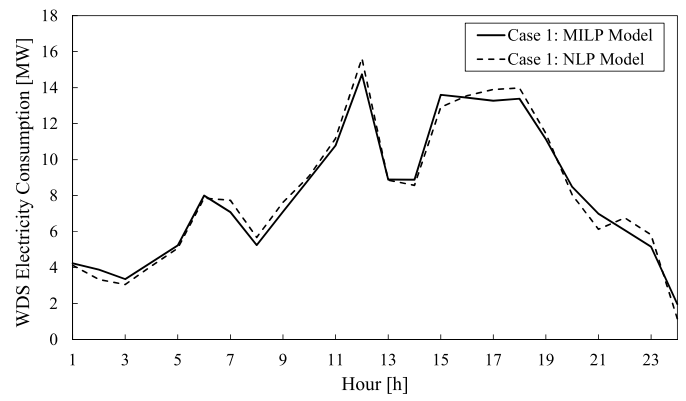


Fig. 10. WDS power consumption using MILP and NLP models for Case 1.

TABLE III  
W-DSOs ENERGY COST USING MILP AND NLP MODELS

| W-DSO Operation Model | W-DSOs Daily Energy Cost (\$) |          |
|-----------------------|-------------------------------|----------|
|                       | Case 1                        | Case 2   |
| MILP                  | 19,959.6                      | 13,874.7 |
| NLP                   | 19,716.7                      | 13,709.9 |

TABLE IV  
COMPUTATION TIME OF W-DSO OPERATION MODELS

| W-DSO Operation Model | Computation time (sec) |        |
|-----------------------|------------------------|--------|
|                       | Case 1                 | Case 2 |
| MILP                  | 28.5                   | 21.6   |
| NLP                   | 449.6                  | 418.1  |

of RAM. The upper bound on the duality gap is set to be zero in the MILP model.

Tables III and IV respectively show the operation cost and computation time of the MILP and NLP models for Cases 1 and 2. In Table III, the original NLP model achieves lower operation costs for W-DSOs in both cases, than the approximated MILP model. More specifically, the additional cost savings achieved by the NLP model amounts for 1.217% and 1.202% of the W-DSO operation cost in Cases 1 and 2 respectively. However, the NLP model takes more than 20 times longer to solve. In Table IV, the MILP model reaches optimal solutions in 28.5 and 21.6 seconds for Cases 1 and 2 respectively, as opposed to the computationally cumbersome NLP model which found a feasible solution after 449.6 and 418.1 seconds for the same cases. Further, the WDS power consumption profiles using the NLP and MILP models are compared in Fig. 10, where the two profiles are very similar and marginally differ over the 24-hours scheduling period. These results indicate that the proposed MILP model provides fairly accurate results at a reasonable computational time.

#### B. Impact of Additional WDS Energy Flexibility

In this study, the number of W-DSOs that are willing to offer their energy flexibility to the power system operator is doubled by adding one more W-DSO to each load bus of the 6-bus system. The total operation cost and total load profile of the 6-bus system in the three cases with more W-DSOs

TABLE V  
OPERATION COSTS OF THE 6-BUS POWER SYSTEM

| Case | Power System Operation Cost (\$) |
|------|----------------------------------|
| 1    | 98,137.0                         |
| 2    | 91,276.2                         |
| 3    | 90,026.0                         |

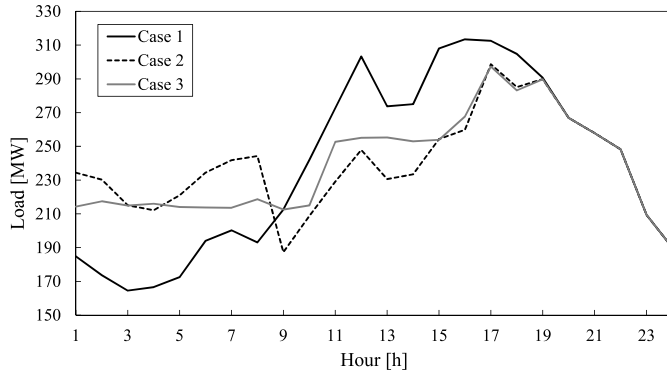


Fig. 11. Daily load profile of the 6-Bus power system with additional WDS energy flexibility.

are shown in Table V and Fig. 11, respectively. In Table V, the total operation cost of the 6-bus system for Case 3 is significantly decreased by \$8,111.0 as compared to Case 1, demonstrating operation cost savings of 8.26% for the power system operator. As expected, when more (six) W-DSOs offer their WDS energy flexibility to the power system operator, a greater peak load reduction is achieved in Fig. 11 for Cases 2 and 3, as compared to the peak load reduction observed for the same Cases in Fig. 8, when only three W-DSOs offer their energy flexibility to the power system. Further, in Fig. 11, the peak load of the 6-bus system experiences a total reduction of approximately 50MW at hour 15 for Cases 2 and 3 as compared to Case 1. However, the load profile of the 6-bus system in Case 3 (where the WDS flexibility is coordinated with the power system operator) is considerably smoother as compared to Cases 1 and 2, providing more cost savings for the power system operator as in Table V. These results demonstrate that the additional WDS energy flexibility leads to much greater power system operation cost benefits, especially when it is optimized by the power system operator.

## VI. CONCLUSION

This paper proposed a model for integrating the energy flexibility of water distribution systems in power systems day-ahead operation. An energy water distribution system operator (W-DSO) is considered who optimizes the operation of WDS flexible components for minimizing its energy costs, while respecting the hydraulic operation constraints of the WDSs. A computationally efficient mixed-integer linear programming (MILP) model is proposed that linearizes the inherent nonlinearities of the WDS operation model, and efficiently solves the proposed problem. The W-DSOs, then, utilize the proposed WDS energy flexibility model in order to calculate and offer

the available energy flexibility to the power system operators, who optimize final decisions on WDS energy flexibility contribution in power systems operation.

The proposed WDS optimization model is implemented on test 15-node WDSs that are supplied by a 6-bus power system. The simulation results indicate that implementing the proposed model significantly reduces the energy cost of W-DSOs. In addition, integrating the WDS energy flexibility reduces the operation cost of the power system, reduce the peak electricity demand, smooth out the load profile, and reduce the electricity prices. These benefits are maximized when the available WDS energy flexibility is optimized by the power system operator. The studies also show that the benefits for power systems are more prominent when more W-DSOs agree to offer and coordinate their energy flexibility with the power system operator. The results indicate that the proposed MILP model has led to computationally efficient solutions that are in acceptable range of the original NLP model solutions. Future works may include expanding the proposed model to a stochastic optimization model, which would account for the uncertainties in the wholesale energy prices, as well as the uncertainty of water demand forecast of WDSs.

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